Md Shamiul Alam Upom

Back-End Developer | Node.js, Python, Payments & Authentication
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SUMMARY

Back-end developer (Python/Node.js) with **one year** in a **production SaaS environment**, working on **payments**, **authentication/RBAC** and **calendar services**. Focused on clean REST APIs, NoSQL data modeling, Docker, and automated testing (Vitest/Pytest).

SKILLS

Programming Languages: Python, TypeScript, JavaScript
Frameworks: Node.js (Express, Hono), Flask/Quart, Django (basic)
Security: JWT, RBAC, EdDSA, BetterAuth, OTP, OAuth (Google)
Databases: PostgreSQL, MongoDB, Redis
Testing: Vitest, Pytest
DevOps & Tools: Docker, Docker Compose, Coolify, Git, Postman, Drizzle ORM, OpenAPI/Swagger
Other: CalDAV, Mongoose, Monorepo, Payments integration

WORK EXPERIENCE

Kouby | Software Engineering Intern | *06/2026 - 08/2026*

Two-person carpooling (ride-sharing) marketplace, final-year internship; back-end and DevOps focus. Repository

- Built the **back-end API** (Hono, TypeScript) in a **pnpm/Turborepo monorepo**, using **shared Zod schemas** so contract changes fail at compile time across the API and the Next.js and Expo clients.
- Implemented authentication with **BetterAuth**: email/password with mandatory verification, **phone OTP**, **Google sign-in**, admin/user roles, and **cookie and bearer sessions** with expiry and rotation.
- Designed the **PostgreSQL data layer** with **Drizzle ORM** and migrations.
- **Handled DevOps**: wrote **Dockerfiles** for the API, web and mobile apps and a **Docker Compose** stack (Postgres, Redis, MinIO) with a **one-shot migration container** that runs before the API starts, plus **Coolify deployment configuration**.
- Documented the API with **OpenAPI/Swagger**, validated environment config with Zod at startup, covered auth middleware with **Vitest tests**, and contributed to the web and mobile front-ends.

Systemica Institut | Backend Engineer | Remote, Germany | *10/2024 - 09/2025*

Contributed to core back-end services and the internal developer platform of a SaaS product.

- Contributed to a **PSD2-SCA compliant payment system** with **automated SEPA flows**; built the **payments API** with **automated reconciliation** and **payment reminders**.
- Built **CalDAV calendar services** (Express.js, Mongoose, Redis, RBAC) handling **~2.2k RPS**, utilizing **Redis caching** and **index optimization** to offload primary database reads.
- Maintained **JWT authentication with RBAC**: **EdDSA-signed tokens** and **session/refresh-token rotation** to limit the impact of a stolen token.
- Co-built a Node.js **monorepo template** with shared linting, CI, auth and logging scaffolds, cutting new-service setup from **7 days to 1 day (~88% faster)**.

Systemica Institut | Freelance Software Engineer | Remote, Germany | *07/2024 - 09/2024*

- Contributed to a Node.js/MongoDB **ticketing service**, load-tested to sustain **millions of emails/day** at **p95 = 150 ms**; **auto-deduplication** projected to cut SLA from **4 h to 1.5 h**.

PROJECTS

MikroTik RouterOS Management | Python, Flask/Quart, MongoDB | GitHub
Back-end and front-end to operate RouterOS/PPPoE for ISPs.

Okaz Marketplace | PHP, HTML, CSS | GitHub
REST API with a minimal front for a small marketplace.

EDUCATION

KEYCE Academy | BSc in Computer Science | *03/2024 - 06/2026 (results pending)*

LANGUAGES

English – Fluent • **French** – A2 in progress • **Bengali** – Native